

[FINA] Digital Wallet Document

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1. Welcome to e-Wallet Documentation

Digital Wallet Platform is a central hub, connecting seamlessly with various components to offer both end users and business owners quick and easy access to a vast array of financial services.

Our user-friendly mobile app and web portal put convenience and control at your fingertips. No matter where you are or when you need it, Digital Wallet empowers you to manage your finances effortlessly.

Ready to unlock financial freedom?

Getting started with Digital Wallet is a breeze! Simply follow our comprehensive [Getting Started guide](#).

Need a helping hand?

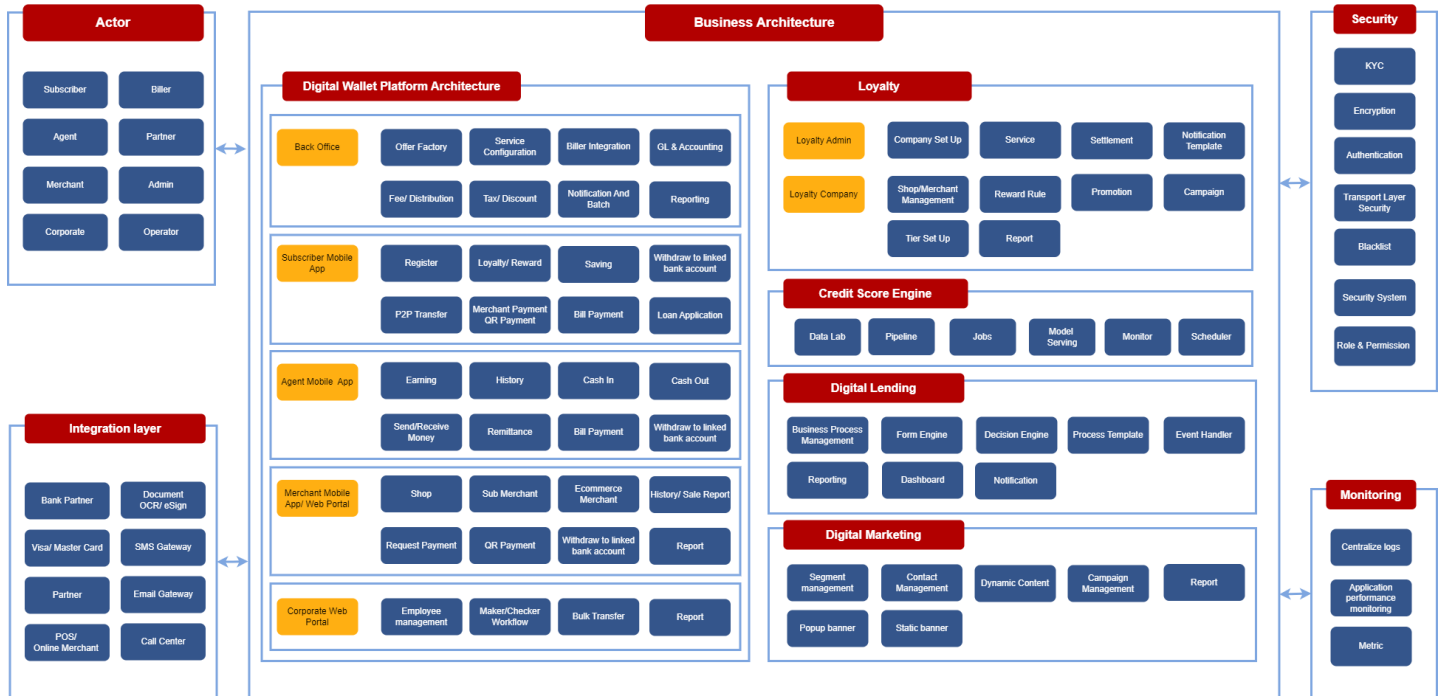
Digital Wallet offers a wealth of resources to ensure a smooth experience:

- [End User Documentation](#): Dive deep into every feature with detailed explanations and step-by-step instructions.
- [Developer Documentation](#): Unleash your potential. This documentation provides comprehensive details about APIs, Webhooks, and everything you need for development.
- **Knowledge Base (backlog)**: Get answers fast! Our knowledge base is packed with tutorials, FAQs, and how-to articles to address any queries you may have.

2. Overview

The platform helps you create a strong marketing strategy for your business. With e- Wallet, you can:

2.1. Components



2.2. Participants

Types of end users on the digital wallet platform are listed below:

User	Definition
Subscriber	Subscribers are the primary users of the digital wallet platform, who have registered and are using an e-wallet service to conduct financial transactions and manage personal finances
Agent	- Agents are representative entities that act as intermediaries between the digital wallet platform and subscribers. - They are involved in conducting transactions or providing services within an e-wallet system.
Merchant	- Merchants are businesses, stores, or services that accept payments through the digital wallet platform - They can receive payment, refund transactions, and track their earnings directly on application
Corporate	Corporates are business users, who use our platform to efficiently manage their employees through bulk transfer capabilities and a robust maker-checker workflow

Officer	Depending on each financial entity, officers are divided into: • Operation: Manages day-to-day operations to ensure smooth and efficient procedures • Business/ Product Owner: Determines product features, functionalities, and priorities based on market needs and business goals. • Accountant: Manages financial records, transactions, and reports to ensure accurate and transparent accounting. • Call Center: Addresses customer inquiries, resolves issues, and guides users through various processes. • Developer: Implements features, functionalities, and technical solutions according to business requirements. •
Biller	• Each biller represents a bill payment service, such as electricity, water, gas, mobile phones, internet, and television,... • Through the digital wallet platform, subscribers can conveniently pay their bills, such as utility bills, rent, or other recurring payments.
Aggregator	Aggregators are utility companies or service providers that provide or refer many bill payment services to e-wallet system
Partner	Partners are the various third-party services that connect with our platform, such as banks, SMS providers, and MPU providers.

2.3. Channel

Channel	Description
Subscriber Application	An application used by customers that allows them to effortlessly link payment methods, make payments, transfer money, and conveniently monitor transaction history,...
Agent Application	Provide agents with services like cash in/cash out, bill payment, transfer money,... on behalf of customers
Merchant Application	Merchants can use merchant application to receive payment, track their earnings, refund payment, enhancing merchant management.
Admin Portal	A web-based platform for administrative users within the financial entity, that enables administrators to manage user accounts, monitor system performance, and configure service and others settings.
Corporate Portal	A portal tailored for corporate clients or businesses that use Digital Wallet Platform. They can utilize Corporate Portal for employee management, streamlined bulk transactions, and ready access to pertinent financial reports
Biller Portal	A centralized portal used by billers or service providers to manage billing and payment processes.

3. Key Concept

Channel	Group	Key Concept	Description
Back office	User	Customer	- Refers to individuals who utilize the platform for personal financial management purposes. - Their services include: - Transfer to wallet - Bill Payment (Utility Bill) - Mobile Operator ...
		Agent	- Refer to individuals who assists customer transfer money, bill payment,... to earn commission - Commission will be decided by business owner and agents
		Merchant	- Merchant can be either an individual or an organization, who uses Digital Wallet as payment method for their transactions - Their services include: - Request payment from Subscriber - Receive money from Subscriber - Refund money to Subscriber - Point Settlement for Subscriber
		Biller	- Refer to an entity that sends bills to users for payment electronically through the platform. - These bills can come from various sources, and users can conveniently pay them directly using their digital wallet balance
	Product Setup	Transaction Design	- Refer to docs Transaction Design - Used to design components of a transaction - Transaction Definition: define GL flow - Transaction Validation: define all the rule to validate transaction - Transaction Fields: define fields and rules to validate those fields

		Offer Factory	Offer Factory empowers you to create and manage targeted offerings for different client segments. This powerful tool simplifies the process, allowing you to define everything from: • Pocket Limits: Set spending limitations for specific offers. • Services: Define the core functionalities included in each offer. • Notifications: Configure how clients are notified about their offers. • Fees & Taxes: Clearly outline any associated fees and taxes for each offer. • Authentication: Set authentication requirements for accessing different offers.
		Fee/ Distribution Matrix	The Fee/Distribution feature equips business owner, with the power to create and manage fee structures that perfectly align with your unique needs. Our flexible fee module offers three distinct options: • Basic Fee: This option combines a fixed fee with a percentage-based charge, offering a simple and straightforward approach. • Advanced Fee: Design your own custom fee structure to fit your specific business model and pricing strategies. • Individual Fee: Business can set unique fees for each client. This caters to personalized offerings and specific client needs.
		Tax	
		Notification	• Notification feature offers business owner ability to set up notification template. • Business can customize receivers, method (sms, push noti, email), type of notification (transaction, promotion, system),...
		Error message mapping (in progress)	• Offer business owner with a brilliant tool to create and manage multi-language error message directly on Admin Portal • The language of error message is based on Application language

	Biller Integration	Biller Integration (Biller Form/Category)	• Biller Form: set up dynamic biller form without the need to do coding • Biller Category: categorize Billers into groups to help user easily navigate and use
		Home Icon	• Home icon on Subscriber Application can be set up in this section. • It is not used for Agent and Merchant Application
	Accounting	COA	• A structured list of all the financial accounts or categories used by a business or organization to record its financial transactions • can be changed according to requirement
		Journey Entry	• Record of a financial transaction or event made in a company's general ledger • Each Journal entry includes 1 Debit and 1 Credit Account, describing money flow from/to within system
		Transaction 360	History of all the transactions updated real-time, can be used to check and analyse error (if have)
		Settlement	
		Reporting	• Report sections provide insights for business owner. • Standard reports: ○ Transaction Detail ○ Transaction Summary ○ GL Details ○ ...
	Workflow		• Concept of this section: maker and checker to create and approve requests ○ Workflow Definition ○ Workflow Config -> used to set up Workflow, just like Transaction design ○ New Request: used to create new request. ○ My Request: If user is maker → he/she can check his request's status in this section ○ My Task: If user is checker → he/she can check and approve/reject request in this section ○ Report/Dashboard • Some commonly used workflow: ○ Reversal transaction: used to reverse transaction

			○ Adjustment transaction: used to make adjustment when there are any errors/mistakes in accounting ○ Operator Cash in/out: operator can help cash in/out money for client
Subscriber App		Onboarding	Two ways Subscriber can register • Subscribers do self-register on Application, then operator approve on Admin portal • Subscriber can be onboarded by Operator on Admin Portal
		KYC Upgrade	• Seft- upgrade KYC with NRIC, Live Photo,... • Status of approval will be notified on application
		Send Money • To wallet user • Non-wallet user	• To wallet user: to transfer money to customer (receiver) who also uses wallet application. Customer (receiver) will receive money directly on application. • Non-wallet user: to transfer money to customer (receiver) who DOES not use application. Receiver will go to Agent to cash out money
		QR scan	• Customers scan other customer's QR to transfer money, applied for wallet user only
		Request money	• Request money from other wallet user and receive money directly on application
		Cashin via • Bank • Agent • Visa	• Cash in = Topup money to wallet. • Method of cash in ○ Cash In money by Bank account: user must link bank account to use this feature ○ Cash In money by Agent: user goes to agent, give agent money and agent will help customer cash in money ○ Cash In money by Visa: cash in money from Visa
		Cashout via • Bank • Agent	Cash out = Withdraw money from wallet • Bank account: user must link bank account to use this feature • Agent: user goes to agent to withdraw money

		Bank Linkage	Customer can utilize this feature to link bank and cash in/out money via bank
		Cashcode	Cash code is used with transaction like: Cash out via Agent, Send money to non-wallet
		Static QR/ Payment QR	- Static QR: - used for request money, transfer money - can specify amount - can be saved to device for further use - Dynamic QR/Payment QR Code: - used for Subscriber Pay Merchant service - will expire after 120s
Agent App		Onboarding	Agent will be onboarded by Operator through Admin Portal
		Transaction	- Cash In: Agent can assist customer in cashin (topup) money to their wallet - Cash Out: - Request Cash Out: Simply understand as Customer request cash out money from their own wallet - Cashout by Cashcode: Simply understand as Customer sends money to a non-wallet user - Send/Receive Money: In case both Sender and Receiver don't have application, they can still go to Agent to ask for Send/Receive money. Customers will need to provide Agent information like: - Bill Payment: Agent can support Customers in making bill payment for electricity, water, internet,...
		Earning	Record earning of each transaction Agent has carried out For example: when Agent cash out money 1000 for customer, they earn 100
Merchant App/ Portal		Onboarding	Merchant will be onboarded by operator on Admin Portal

		Shop/ Employee	• One Merchant can have multiple shops and employees. • Merchants can effortlessly manage and monitor the performance of all their shops from a single, user-friendly application. • Merchants have the power to define and assign permissions for each role within their organization. This promotes security and allows for customized access based on employee responsibilities.
		Transaction	• Request Money: Easily initiate a payment request to a customer. Specify the amount and any relevant details within the application. → Subscriber approve/reject on Subscriber Application • Scan QR: Scan Subscriber's QR to request payment • Refund Payment: Process and issue refunds to customers with ease
Corporate Portal		Onboarding	• Corporate user onboarding will be handled by Operators within the Admin Portal.
		Process	• These pre-built processes offer a convenient starting point, saving you valuable time and resources. ◦ Bulk Transfer to Wallet: Effortlessly move funds to multiple wallets simultaneously. - Bulk Transfer to Non-Wallet: Efficiently transfer funds to accounts outside your platform. ◦ ... • Our platform empowers corporations to personalize the maker-checker process for each standard process
		My task/ My Item	Our platform adopts a secure and reliable "maker-checker" workflow, featuring two distinct user roles: • Maker: This user creates, modifies, or submits items (tasks, requests, documents) for approval. • Checker: This user reviews and approves (or rejects) items submitted by makers.

4. Quick Start Guide

Digital Wallet is a comprehensive **financial management solution**. It caters to a wide range of users, from individuals seeking to simplify their personal finances to businesses aiming to streamline transactions. This user-friendly guide provides a quick walk-through to getting started with Digital Wallet. By following a sample scenario, you'll learn how to navigate the platform's features and unlock its full potential for managing your money effectively.

	Scenario	Step-by-step
1	Create Officer Account	1. Login to Digital Wallet Admin 2. Access Operations → Officer 3. Create new officer account by filling in: ○ Employee ID ○ Full name ○ Phone ○ Email ○ Group 4. After successfully creating, account information will be sent to officer's email
2	Setup COA, Currency	1. Login to Digital Wallet Admin 2. Access Accounting → Chart Of Account 3. Create new account by filling in: ○ Code ○ Name ○ Type: Asset, Liability, Equity, Income, Expense ○ Parent: only if this account is under a parent account 4. After successfully creating, new account will be updated and officer can immediately use it to config GL Flow
3	Setup User group and user role	1. Login to Digital Wallet Admin 2. Access System Config → User Role 3. Officer can ○ choose User role from dropdown list ○ create new User role 4. After choosing User role, officer can choose access permission simply by ticking/unticking in the box

4	Onboard new subscriber, Agent. Merchant	1. Login to Digital Wallet Admin 2. Access Operations → Subscriber → Create 3. Officer has 2 options ○ Quick registration: only fill in basic information like Full Name, Phone, Birth, Gender After that, customer can self KYC on their application ○ Full Registration with KYC: onboard with more information like: Portrait Image, NRIC Information (Image, Number) or Passport Information (Image, Number) 4. After successfully creating, subscriber will receive their account information via SMS or email
5		1. Login to Digital Wallet Admin 2. Access Operations → Merchant → Create 3. Officer gathers Merchant information and keys in all required information 4. After successfully creating, merchant will receive their account information via SMS or email
6		1. Login to Digital Wallet Admin 2. Access Operations → Agent → Create 3. Officer gathers Agent information and keys in all required information 4. After successfully creating, merchant will receive their account information via SMS or email

Setup Subscriber transfer to wallet Transaction with information as follow

- Service name: Transfer to Wallet
- Fee: 25 for sender
- Distribution: 100% to system
- Apply for Subscriber Level 2

1. Set up Service
<https://digitaltransformation.atlassian.net/wiki/spaces/FAQ/pages/5059469>
 - Access Admin Portal → Product Management → Service → Create
 - Fill in all required information. You can see example in the image belowSome important information you should pay attention to: Code, Name, Amount Formula

2. As this transaction has fee, access Product Management to configure Fee. Remember to activate Fee after you have done creating it.
Explanation: Debit Fee is Sender Fee, Credit Fee is Receiver Fee

Create Fee

Name: Transfer to Mobile Fee
 Currency: USD
 Description: Fee applied when user do transaction

Tiers

Tier 1

amount > 0
 standard
 percentage * amount > fixed

Percentage: 25
 Fixed Value: 0.00

Buttons: + Create, IF Save, +

Fee

Search for...

#	Code	Name	Description	Created	Created At	Last edit	Editor	Status	Action
1	KT07050402	Transfer to Mobile Fee	Fee applied when user...	SYSTEM USER 00	2024-03-08 17:25	2024-03-08 17:25	SYSTEM USER 00	Active	Detail, Copy, Approve, Delete
2	KT07050407	Mobile Fee	Mobile	SYSTEM USER 00	2024-03-08 17:25	2024-03-08 17:25	SYSTEM USER 00	Active	
3	KT07050404	Fee Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-04 18:04	SYSTEM USER 00	Active	
4	KT07050408	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
5	KT07050409	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
6	KT07050407	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
7	KT07050407	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
8	KT07050407	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
9	KT07050407	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	
10	KT07050407	Request Push Payment		SYSTEM USER 00	2024-03-12 18:10	2024-03-12 18:10	SYSTEM USER 00	Active	

- After creating Fee, go to Product Management → Distribution. In this example, 100% fee will go to System Owner

Create Distribution

Name: Transfer to Mobile Fee Distribution
 Currency: USD
 Distribute From Fee Type: Select Fee Transfer To Mobile Fee

Buttons: + Create, IF Save, +

System Owner	Value
System Owner	100
Offboard Agent	0
Transaction Agent	0
Mobile Agent	0
Merchant	0
Aggregator	0
Distributor	0
Offboard Mobile Agent	0
Cardback to Customer	0
Cardback to Beneficiary	0

Buttons: + Create, IF Save, +

Donut Chart: 100% System Owner

- After that, go to Product Management → Transaction Design → Action → Detail on the service you have just created to add more information for this service

DIGITALWALLET Admin

Services > Transaction Design

Transaction Design

Search For...

Count: 227

#	ID	Code	Name	Status	Version	Created At	Creator	Last edit	Editor	Action
1	6629463080504646704	TRANSFERORWALLETTEST	Transfer to MicroPay - Test	ON	1	08-05-2024	SYSTEM USER 03	08-05-2024	THAO	Action
2	6629623080504646704	VALIDATEGPPROCMPTPAY	Validate QR PromptPay	ON	1	07-05-2024	THAO	07-05-2024	THAO	Detail
3	6629623080504646704	VALIDATEGPPROCMPTPAY	Validate PromptPay ID	ON	1	07-05-2024	THAO	08-05-2024	THAO	Export
4	6629623080504646704	PROCMPTPAYOUTTAGLONUS	TAGLONUS PROCMPTPAY_OUT TAGLONUS	ON	1	07-05-2024	THAO	08-05-2024	THAO	Action
5	6629623080504646704	PROCMPTPAYOUTTAGLONUS	TAGLONUS PROCMPTPAY_OUT TAGLONUS	ON	1	07-05-2024	THAO	08-05-2024	THAO	Action
6	6629623080504646704	PAY_MERCHANT_PAY	Pay Merchant Thai Paboon	ON	1	08-05-2024	SYSTEM USER 08	08-05-2024	SYSTEM USER 08	Action
7	6629623080504646704	PAY_MERCHANT_PAY	Pay Merchant Muech Paboon	ON	1	08-05-2024	SYSTEM USER 08	08-05-2024	SYSTEM USER 08	Action
8	6629623080504646704	SAVECRAFTTEXTINFO	Save Craft text info	ON	1	25-04-2024	SYSTEM USER 01	25-04-2024	SYSTEM USER 01	Action
9	6629623080504646704	ADDUSERTOBLOCKLISTDOCUMENT	ADD user to blocklist and report document	ON	1	04-03-2024	SYSTEM USER 01	22-04-2024	SYSTEM USER 01	Action

- On Tab Transaction Definition, set up GL Flow. Remember to Save tab before moving to the next one
 Example: Customer A transfer 100 to Customer B, Fee = 25
 TransOriginAmount = 100
 TotalCreditFee = 25, distributed 100% to system
 -> Receiver will get TransOriginAmount - TotalCreditFee = 75

DIGITALWALLET Admin

Transfer To MicroPay - Test

Transaction Definition

Step 1

From: [Select User] To: [Select User] Amount: [100] Currency: [THB]

Step 2

From: [Select User] To: [Select User] Amount: [25] Currency: [THB]

Step 3

From: [Select User] To: [Select User] Amount: [75] Currency: [THB]

- Next to tab Transaction Validation is where you can add validation for transaction. Example: validate user/balance/....

DIGITALWALLET Admin

Update Transaction Design

Transfer To MicroPay - Test

Transaction Validations

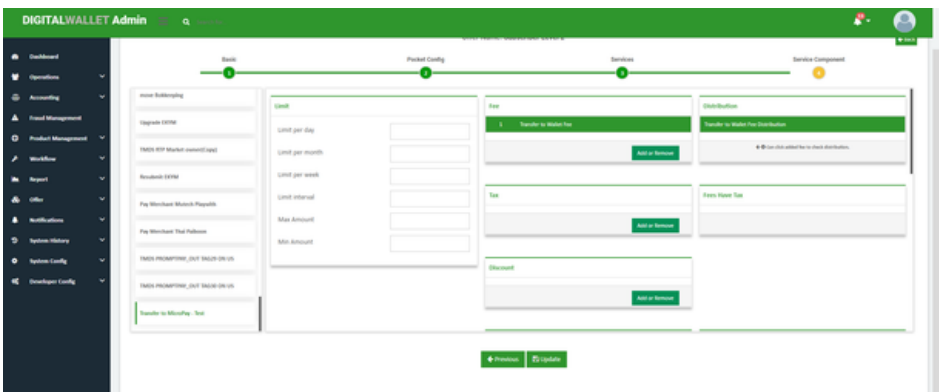
#	Rule Code	Rule Function	Agent	Error Code	Error Message	Status	Action
1	validateSenderAccount	validateSenderAccount	SENDINGPHONE-SENDERCLIENT-SENDING	9001	Sender is invalid	ON	Action
2	validateSenderAccountBalance	validateSenderAccountBalance	SENDINGPHONE-SENDERCLIENT-SENDING	9002	Your wallet balance is not enough	ON	Action
3	validateReceiverAccount	validateReceiverAccount	RECEIVINGPHONE-RECEIVERCLIENT-RECEIVING	9003	Receiver balance exceed limit	ON	Action
4	validateTransferAmount	validateTransferAmount	SENDINGPHONE-SENDERCLIENT-SENDING	9004	Amount too big	ON	Action
5	validateTransferAmount	validateTransferAmount	SENDINGPHONE-SENDERCLIENT-SENDING	9005	Receiver need to be another account	ON	Action
6	validateTransferAmount	validateTransferAmount	SENDINGPHONE-SENDERCLIENT-SENDING	9006	Can not access to service	ON	Action
7	validateRequestCode	validateRequestCode	SENDINGPHONE-SENDERCLIENT-SENDING	9007	Request is paid	ON	Action
8	validateTransferAmount	validateTransferAmount	SENDINGPHONE-SENDERCLIENT-SENDING	9008	Amount too small	ON	Action
9	validateCurrency	validateCurrency	SENDINGPHONE-SENDERCLIENT-SENDING	9009	Currency is not valid	ON	Action
10	validateDailyLimit	validateDailyLimit	SENDINGPHONE-SENDERCLIENT-SENDING	9010	Amount exceeded daily limit	ON	Action

- Next one is Transaction Fields where you configure all the data fields (regex, min/max length, error code, error message,...) for this transaction. You can view example below

The screenshot shows the 'DIGITALWALLET Admin' interface. The main heading is 'Transfer To MicroPay - Test'. Below it, there are tabs: Overview, Transaction Definition, Transaction Validation, **Transaction Fields**, Input Building, Action Result, and Versions And Updates. The 'Transaction Fields' tab is active, displaying a table with 12 columns: S, Order, Field name, Field format, Min length, Max length, Regex, Need format, Required, Status, Error Code, Error Message, and Action.

S	Order	Field name	Field format	Min length	Max length	Regex	Need format	Required	Status	Error Code	Error Message	Action
1	1	REFERENCE	string	24	24		☐	☒	☒	9011	Can not access to card	Action
2	2	OFFERED	string	24	24		☐	☒	☒	9002	Wrong Offer id format	Action
3	3	AMOUNT	float	3	38		☐	☒	☒	9013	Wrong Amount format	Action
4	4	CURRENCY	string	3	3	^[0-9A-Z]{3}\$	☐	☒	☒	9012	Wrong Currency format	Action
5	5	DEVICEID	string	1	254		☐	☒	☒	9017	Wrong Deviceid format	Action
6	6	SENDERPHONE	string	9	33	^[0-9]{9,15}\$	☐	☒	☒	9004	Wrong Sender Phone	Action
7	7	SENDERCLIENT	string	3	33	^[a-zA-Z0-9]{3,33}\$	☐	☒	☒	9005	Wrong Sender Client I	Action
8	8	SENDERID	string	24	24		☐	☐	☒	9006	Wrong Senderid format	Action
9	9	RECEIVERPHONE	string	9	33	^[0-9]{9,15}\$	☐	☒	☒	9008	Wrong Receiver Phone	Action
10	10	RECEIVERCLIENT	string	1	33	^[a-zA-Z0-9]{1,33}\$	☐	☒	☒	9009	Receiver's account i	Action
11	11	RECEIVERID	string	24	24		☐	☒	☒	9007	Receiver's account i	Action
12	12	REQUESTED	string	24	24		☐	☒	☒	9015	Wrong Request id format	Action
13	13	MESSAGE	string	3	255		☐	☐	☒	9016	Wrong Message format	Action
14	14	SCODE	string	3	33		☐	☐	☒	9000	Wrong SCODE	Action

- Next, create Notification template in Notification Template. Notification can be set to be sent via SMS, PUSH or EMAIL and will be sent immediately after customer has done transaction

	
	NOTE: You can apply these steps to make configure other types of transactions

5. Advance Scenarios

	Scenario	Step-by-step
1	Unlock Account	1. Login to Digital Wallet Admin 2. Access Operations → Officer/Subscriber/Agent/Merchant 3. Click Action → Active 4. After successfully Active account, Officer/Subscriber/Agent/Merchant can log into and use application/portal normally
2	Transaction Configuration end to end	1. Create Service: Service 2. Set up Transaction Design: Transaction Design 3. Set up Fee & Distribution: Fee Distribution 4. Set up Discount and Tax (if needed) Discount 5. Set up Notification: Notification template 6. Add to Offer Factory: Offer Factory
3	Online Merchant Onboarding	Online Merchant Onboarding Guide
4	Special Fee	Fee